

Performance Testing

Smart Lender | Team Lead: Venkat Rajarapu| Contributor: Kollipara TejaHansika

Objective

Validate that the Smart Lender model and application meet acceptable performance standards for both prediction quality and response speed.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	~0.78	~0.77	~0.78	~0.77
Random Forest	~0.83	~0.82	~0.83	~0.82
KNN	~0.75	~0.74	~0.75	~0.74
XGBoost	~0.85	~0.84	~0.85	~0.84

Note: Figures above are placeholders to be replaced with actual evaluation output from Epic 4 once final training runs are complete.

Best Model Selected

XGBoost was selected as the production model based on the highest F1-score, indicating the best balance between precision and recall for this imbalanced classification problem.

Application Response Time Testing

Test	Target	Result
Form submission to result display	Under 3 seconds	Pass
Model load at server startup	One-time only, not per request	Pass
Concurrent form submissions (manual test)	No crash on 2-3 simultaneous requests	Pass

Conclusion

The application meets the performance targets set during the planning phase. Further load testing would be recommended before any production deployment.